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Regularities of Rock Destruction with a Single Blow

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Abstract

The operating conditions of the picks of impact breaking and chipping machines are analysed. A method and a test rig developed by the authors for studying the penetration of a pick into rock and the impact-abrasive wear of its material are described. Using freely falling strikers (D = 6 mm, L = 25 mm, conical point with an angle of 30°) made of steels 38XM (AISI 4140) and 55C2 (AISI 9255), which differ in strength properties and wear resistance, impacting fragments of granite with a strength f = 6.2 on Prof. M.M. Protodyakonov's scale, the regularities of rock destruction by a single blow (v_0 = 0.91 m/s, A_0 = 1.16 J) were investigated, taking into account the blunting of the striker tip. On the basis of the principles of the two-element classical impact system, a mathematical description of the penetration of the striker into rock under the experimental conditions is proposed. The calculation results, confirmed by experimental data, show that with a single penetration of a cone-shaped impact tool into granite, the penetration depth decreases with increasing diameter d_K of the blunt area of the striker according to a logarithmic relationship. The calculated numbers of blows after which the critical value d_K^* = 2.45 mm (≈ 2.3 mm according to experiment) is reached and penetration of the striker into the rock ceases are 124,552 blows for steel 38XM and a 15 times greater number of blows (1,849,580) for the harder steel 55C2.

The authors conclude that the service life of picks and the productivity of impact breaking and chipping machines can be substantially increased by manufacturing picks from more wear-resistant steels or by applying methods that increase the hardness of the head part of the tool, for example cold treatment.

Keywords: *impact tool, rock, penetration depth, striker material, impact-abrasive wear, blunt area.*

Introduction

Impact breaking and chipping machines are widely used in driving, enlarging and scaling of mine workings and tunnels, for dismantling refractory linings during repair of blast and open-hearth furnaces, converters, steel-pouring and hot-metal ladles, in excavation of frozen and rocky soils, opening of road pavements, demolition of concrete, reinforced concrete and brick structures, etc. [1]. The working tool of these machines is usually a cylindrical pick in the form of a cone or wedge, sharpened at an angle of, as a rule, 30 to 40° (Fig. 1). This reduces the frontal resistance to penetration and, owing to friction self-locking, enables the tool to remain stationary relative to the material and to maintain the stressed state in it in the intervals between blows [2].

[Figure 1 — see original PDF, page 1] Working element (pick) of the DAEMO B180 hydraulic breaker

The breaking process with such machines begins with penetration of the tool into a mass having free surfaces fairly close to each other, to such a depth that the tensile and shear stresses arising from the penetration tear off a piece of the required size from the mass. The working capacity of the machine is determined by its ability to drive the impact tool to a given depth, and the productivity indicator is the rate of penetration [2].

In practice, the machine operator selects a point on the surface of the rock fragment to be broken, onto which he delivers a series of blows at the required angle (from 0 to 45°). If a piece of rock does not spall off within the impact time specified by the machine's operating instructions, the point of impact is moved to another point on the surface to prevent overheating of the impact tool. For example, for the Hyundai R-170W-7 excavator equipped with a mounted DAEMO B180 hydraulic breaker, this impact time is no more than 30 s. According to the operating instructions of the hydraulic breaker, using the impact tool as a lever when breaking rock is not permitted.

During operation of such machines, impact-abrasive wear of the pick material produces a blunt area in its pointed part, whose size grows as the pick is used. This leads to an increase in the contact area between the tool and the rock upon impact and a decrease in the intensity of impact breaking. As a result, from a certain moment, penetration of the pick into the rock practically stops, and replacement or resharpening of the pick is required to resume it. As experience with the above-mentioned DAEMO B180 hydraulic breaker shows (impact energy 2.35 kJ; pick mass and dimensions M = 93 kg, L = 1165 mm, D = 120 mm; length of the pointed part 200 mm), for this machine this moment occurs when the pointed part of the pick is worn by approximately 30%. Consequently, a pick lasts for 4...5 resharpenings. Considering that replacement of the massive pick and its resharpening are very labour-intensive processes, the development of methods that increase the resistance of hydraulic breaker pick materials to impact-abrasive wear and, consequently, the service time of picks before resharpening is a topical task.

In the present work, under conditions approaching the operating conditions of the picks of impact breaking and chipping machines, using as an example granite with a strength, determined according to the standard [3], equal to 6.2 on Prof. M.M. Protodyakonov's scale, the regularities of rock destruction by a single blow of a freely falling striker were investigated. The strikers were made of steels (38XM and 55C2) used for manufacturing hydraulic breaker picks, which differ substantially in hardness and wear resistance. The chemical composition of the steels, the heat treatment modes used in manufacturing the strikers, the main physical and mechanical properties and the resistance to abrasive wear, established experimentally by the authors earlier [4], are given in the table.

Chemical composition and main physical and mechanical properties of the striker materials

Steel grade	Content of main elements, %	Heat treatment	Parameter values				
			$\sigma_{0.2}$, MPa	σ_B, MPa	KCV, J/cm^2	Hardness, HRC	Rw, N·m/mm^3
38XM	0.39C, 0.19Si, 0.79Mn, 0.98Cr, 0.15Mo	Quenching from 870 °C + tempering at 200 °C	1478	1933	40.6	51.1	69.0
55C2	0.53C, 1.54Si, 0.73Mn, 0.04Cr		1887*	2468*	5.6	59.6	97.0

* Because of brittle fracture of the specimens over a wide range of stresses, the values of $\sigma_{0.2}$ and σ_B were calculated from the hardness of the steel using the formulas from [5].

Experimental method

The experiments were carried out on a rig similar to that described in [6, 7]; its appearance and schematic diagram are shown in Fig. 2. Striker 2 (a cylinder with diameter D = 6 mm and length $l_{уд}$ = 30 mm with a conical point with apex angle β = 30°), made of one of the steel grades indicated above, was weighed on an analytical balance (weighing accuracy ± 0.001 g) and then fixed vertically in the massive ram 3 (cylinder D = 50 mm, L_6 = 293 mm) of the rig. The total mass M of the striker and ram was 2.8 kg. A cylindrical rock fragment 1 (D = 58 mm, H = 46 mm) was placed under the striker at a fixed distance using a stationary holder. By means of electric motor 7, gearbox 6, an intermediate shaft with cam 5 and a spindle with roller 4, the ram with the striker was raised to a certain height, from which they fell onto rock fragment 1, the pointed part of the striker delivering a blow to a given point on its surface. The drop height H of the striker and ram was 42 mm; their pre-impact velocity v_0 and kinetic energy A_0 were 0.91 m/s and 1.16 J.

The striker was weighed again and its length was measured (micrometer, measurement accuracy ± 0.01 mm) to determine the mass loss $\Delta m_{уд}$ and the length $\Delta l_{уд}$ of its worn part, from which the diameter $d_{уд}$ and area $f_{уд}$ of the resulting blunt area were calculated.

For the rock, the shape of the spall crater was analysed, and its depth $h_л$ was measured using a dial indicator (measurement accuracy 0.01 mm). For some craters, $h_л$ was measured again after removal of the fracture products.

[Figure 2 — see original PDF, page 3] Rig for studying impact destruction of rock and impact-abrasive wear of the striker material (a) and its schematic diagram (b): 1 — rock fragment; 2 — striker; 3 — ram; 4 — roller; 5 — cam; 6 — gearbox; 7 — electric motor

The subsequent tests (39 blows) were carried out with the same striker, changing the point of impact on the fragment with each blow, measuring the parameters $\Delta l_{уд}$, $\Delta m_{уд}$, $h_л$ and determining $d_{уд}$. For each test interval, the total mass loss of the striker $\Sigma\Delta m_{уд}$ was calculated. The dependences of $\Delta l_{уд}$, $\Sigma\Delta m_{уд}$, $d_{уд}$ and $h_л$ on the number of blows n were plotted.

Similar tests were carried out with a striker made of the other steel grade.

In a special series of experiments, by successively treating the striker tip with an abrasive, the diameter of the blunt area was increased up to the critical value $d_{уд}$*, starting from which the impact of the striker on the rock no longer produced a crater.

Experimental results

As observations showed, practically no rebound of the striker is observed during impact on granite. The result of impact for strikers of both steel grades is the appearance of circular blunt areas on the pointed part of the striker, whose diameter $d_{уд}$ increases with the number of blows.

Destruction of granite is characterised by ejection of dust-like fragments from the fracture zone and the formation of spall craters whose shape is close to the configuration of the penetrating part of the striker, but of larger diameter. At the bottom of the craters, a compacted core of powdery fracture products is found, differing in colour (white) from the rest of the rock. The depth of the craters increases approximately 1.5 times after removal of these products. No cracks radiating from the craters along the rock surface are observed (Fig. 3).

[Figure 3 — see original PDF, page 3] Spall craters on the surface of a granite fragment after impacts of the pointed striker

It was established that for both steel grades analysed, the form of the dependences $\Sigma\Delta m_{уд}$, $\Delta l_{уд}$, $h_л = f(n)$ is similar (Fig. 4): the total mass loss of the striker $\Sigma\Delta m_{уд}$ increases at an approximately constant rate (equations of curves 2), which is lower for the harder steel 55C2 (0.11 mg per blow compared with 0.21 mg for steel 38XM), while the loss of striker length $\Delta l_{уд}$ (curves 4) and the resulting crater depth $h_л$ (curves 3) decrease monotonically. Moreover, over the entire range of the number of blows n, the crater depths $h_л$ for steel 55C2 are somewhat larger. The diameter $d_{уд}$ of the resulting blunt area of the striker, calculated from $\Delta l_{уд}$, increases with the number of blows according to a dependence satisfactorily approximated by a logarithmic equation (equations of curves 1).

[Figure 4 — see original PDF, page 3] Dependence of the blunt area diameter $d_{уд}$ (curve 1), total mass loss $\Sigma\Delta m_{уд}$ (curve 2) and length loss $\Delta l_{уд}$ (curve 4) of the striker, as well as crater depth $h_л$ (curve 3), on the number of blows n for strikers of steels 38XM (a) and 55C2 (b). Fitted equations shown on the graphs: (a) $d_{уд} = 0.1499 \ln(n) + 0.6991$; $\Sigma\Delta m_{уд} = 0.2129n + 3.0535$; (b) $d_{уд} = 0.1273 \ln(n) + 0.6208$; $\Sigma\Delta m_{уд} = 0.1108n + 4.1169$. Axes: number of blows n; mg (left); mm (right)

Mathematical model of the impact destruction process and tool wear

In developing the mathematical model of the process, we proceeded from the fact that the striker together with the ram in our case is a structure of comparatively small length. For this reason, it was considered that the influence of wave processes on the impact parameters can be neglected and the interaction between the striker and the material being destroyed can be described by the equations of the two-element classical impact system [2]. As the calculation results showed, the condition for the applicability of these equations,

$$t_m \geq 3T, \quad (1)$$

where t_m is the impact duration ≈ 0.6...1.4 ms; T is the period of natural oscillations of the striker and ram ($T = 2(L_{уд} + L_б)/a$, where a is the velocity of the shock wave in steel ($5.15 \cdot 10^3$ m/s), i.e. T = 0.13 ms; consequently, condition (1) is satisfied).

Based on observations of the impact process and of the structure of the resulting craters, it was assumed that penetration of the striker into the rock is characterised by two main stages:

- elastic deformation of the rock until the stress $\sigma_к$ is reached over the area of its contact with the striker, at which explosion-like destruction of the rock under the contact area occurs;
- advance of the striker through the core of destroyed rock, with part of the products being displaced from the crater and the remainder being compacted, without reaching the bottom of the fracture zones, and the striker being thrown back by the forces of residual elasticity.

The dependence of the rock resistance force N, N, on the deformation a, m (Fig. 5), for this case corresponds to a V-type load characteristic [2] and consists of three force functions: loading N_1 during elastic deformation of the rock under the contact area, loading N_2 during penetration of the striker into the rock fracture products, and unloading N_3, each of which is characterised by its own stiffness (g_1, g_2, g_3, N/m). It was assumed that the functions N_1 and N_2, as for penetration of a body with a constant contact area, are linear, and that the function N_3, owing to the absence of residual

stiffness of the system, is equal to zero. The hatched area of the figure in Fig. 5 is the work A_p expended on destruction.

The maximum rock resistance force N_m and penetration a_m, as well as the impact duration t_m, for such a load characteristic are [2]:

$$N_m = N_p \sqrt{(1 - g_2/g_1 + 2A_p g_2/N_p^2)}; \quad (2)$$

$$a_m = (N_p/g_2) \cdot [\sqrt{(1 - g_2/g_1 + 2A_p g_2/N_p^2)} - 1 + g_2/g_1]; \quad (3)$$

$$t_m = \sqrt{(M/g_2)} \cdot arctg \sqrt{(2A_p g_2/N_p^2 - g_2/g_1)}, \quad (4)$$

where M is the mass of the falling object; N_p is the force of contact destruction of the rock

$$N_p = f_к \sigma_к, \quad (5)$$

where $f_к$ is the contact area; $\sigma_к$ is the stress at the contact point at which the tensile and shear stresses σ_p arising in the rock upon impact lead to its explosion-like destruction under the contact area.

[Figure 5 — see original PDF, page 4] Load characteristic of the system adopted in calculating the impact parameters ($N_1 = g_1 a$; $N_2 = N_p + g_2 a$; $N_3 = 0$)

Assumptions adopted in the calculations:

- the dynamic hardness of the rock (2100 MPa for granite [8]) can be taken as the value of the critical contact stress $\sigma_к$;
- to determine the contact stiffness of the system g_1 for a circular contact area of diameter $d_к$, the formula from [2] is valid:

  $$g_1 = 2E_п d_к / [3(1 - \mu_п)], \quad (6)$$

  (with the elastic modulus $E_п$ = 50 GPa and Poisson's ratio $\mu_п$ = 0.2 for granite [2]);
- by analogy with the stiffness ratio g_2/g_1, which can be established from an analysis of the experimental force characteristics of contact destruction of brittle rocks given in [2], the stiffness g_2 in our case can be taken equal to 0.16 g_1;
- only about half of the kinetic energy A_0 of the striker is spent on elastic deformation and destruction of the rock (the remaining energy is spent on elastic and plastic deformation of the striker, as well as on wear of its contact surface and heat losses to the environment [8]); therefore we take $A_p = 0.5A_0 = 0.58$ J.

Fig. 6 presents the dependences of the maximum rock resistance force N_m and the penetration of the striker into the rock a_m, as well as the impact duration t_m, calculated by formulas (2)–(5) from the value of N_p for the range of contact-area diameters $d_к$ ($f_к = \pi d_к^2/4$) occurring in the experiments. The same graph shows, for the corresponding values of $d_к$, the crater depths $h_л$ determined experimentally (curves 3 in Fig. 4). The arithmetic mean of the striker $d_{уд}$ before and after the blow was taken as the diameter $d_к$ of the contact area between the striker and the rock.

[Figure 6 — see original PDF, page 5] Calculated parameters of a single blow as a function of the contact-area diameter $d_к$: 1 — impact duration t_m; 2 — resistance force N_m; 3 — maximum penetration depth a_m compared

with the experimentally determined depth of the resulting craters for strikers of steel 38XM and 55C2

As can be seen in Fig. 6, at a constant force impulse $Mv_0 = N_m t_m$, with increasing diameter $d_к$ of the contact area ($d_{уд}$ of the striker blunt area), the maximum rock resistance force N_m (curve 2) increases monotonically, while the impact duration t_m and the maximum penetration depth a_m of the indenter decrease. At the same time, as can be concluded from comparing the positions of the experimental points and the curve, the regression line of the dependence $a_m = f(d_к)$ shows satisfactory agreement with the experimental data for both steel grades and can be described by the logarithmic equation

$$a_m = 0.2861 - 0.3090\ ln(d_к). \quad (7)$$

Extrapolation of dependence (7) to higher values of $d_к$ made it possible to obtain the critical value of the blunt-area diameter $d_к^* = 2.45$ mm, at which the maximum penetration depth a_m decreases to the value a_1 (see Fig. 5). With a blunt area of this size, all the energy A_p is spent only on elastic deformation of the rock without its destruction and, in accordance with Fig. 5 and taking into account formulas (5) and (6):

$$A_p = N_p a_1/2 = N_p^2/(2g_1) = 3\sigma_к^2\pi^2 d_к^{*3}(1 - \mu_n) / (64E_n). \quad (8)$$

A value of the critical diameter close to this was obtained experimentally: $d_{уд}^* \approx 2.3$ mm.

The satisfactory agreement obtained between the calculation and experimental results can serve as confirmation of the proposed mathematical model and of the validity of the assumptions made.

Substituting the established value of the critical diameter ($d_к^* = 2.45$ mm) into the experimentally determined equations describing the change in the blunt-area diameter of strikers made of the different steels with the number of blows (equations of curves 1 in Fig. 4) made it possible to calculate the number of single blows after which penetration of the striker into the rock ceases: 124,552 blows for steel 38XM and a 15 times greater number of blows (1,849,580) for the harder steel 55C2.

Conclusion

As shown by calculation and experiment, with a single penetration of a cone-shaped impact tool into granite, the penetration depth depends on the size of the blunt area of the striker and decreases with increasing blunt-area diameter according to a logarithmic dependence. If it is assumed that this regularity also holds for repeated blows at one point under the real operating conditions of impact breaking machines, it can be concluded that manufacturing strikers from harder, wear-resistant steel, as well as applying methods that increase their hardness, for example cold treatment [4], should slow down the growth rate of the blunt area. This, on the one hand, will lead to an increase in the total depth of spall craters obtained for a given number of blows and, on the other hand, will increase the number of blows that the striker is able to deliver before reaching its maximum permissible bluntness; that is, it will help increase the productivity of the impact breaking and chipping machine and extend the service life of its tool before replacement or resharpening.

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